

Yihui Yan

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Education

East China Normal University | School of Software Engineering 2024.09 — Present

Research interests: Integration of Mobile Edge Computing (MEC), Internet of Things (IoT), and Deep Learning.

Xidian University | School of Computer Science and Technology 2020.09 — 2024.06

GPA: 3.8/4.0 | Ranking: 21/332 (Software Engineering) & 4/61 (Network Direction)

Main Courses: Data Structure (99), Computer Network and Communication (91), Compiler Principles (97), Algorithms (99), Advanced Mathematics (90), etc.

Awarded university scholarships for four consecutive years; First Prize in Shaanxi Province Mathematical Modeling Contest; Excellent League Member; Outstanding Graduate.

Project Experience

Huawei Technologies Co., Ltd. | AI Engineer Intern | 2025.07 — 2025.10, Shanghai

Participated in the development of the large model testing platform (**Yuntan**), deploying models on cloud servers, building intelligent benchmarking agents via MCP framework for automated task generation and scheduling, and developing power monitoring components for data collection, analysis, and visualization. Also optimized backend functionalities, **contributing over 2000 lines of code**.

- **Model Deployment:** Deployed inference and training services for Llama and Qwen models on cloud servers based on the **Ascend-vLLM** framework, and performed performance benchmarking to obtain key metrics such as **QPS** and **TTFT**.
- **Performance Monitoring Component:** Built a monitoring pipeline using **Prometheus + Grafana**, and independently developed the **A3-Power** component to visualize and analyze **9 metrics** (power consumption, memory usage, NPU temperature, etc.) across multiple NPU nodes with automated incremental analysis.
- **MCP Agent Development:** Constructed a custom MCP server based on **FastMCP** and integrated with the internal **Lingque (Dify)** platform to achieve automatic generation, scheduling, and cross-network execution of benchmarking tasks, improving testing efficiency—**reducing task execution time from 4 hours to 30 seconds**.
- **Java Backend Optimization:** Implemented complete logic across API, DAO, Common, Mapper, and MySQL layers; completed front-back-end API integration and Mapper-level unit tests to ensure system stability and correctness of permission and query functions.

“Little Detective” Device Maintenance Management System | 2022.11 — 2023.06

- Led the development of a **SpringBoot + Vue3** based front-back-end separated system supporting device registration, maintenance requests, and repair management;
- Designed the overall architecture and API specifications, improving system scalability and maintenance efficiency.

Intelligent Glass Composition Analysis and Classification System | 2021.08 — 2021.12

- Applied the **Fisher Discriminant Method** for classification of different glass compositions;
- Used **Fuzzy Clustering Algorithms** to identify subcategories and analyze variations between pre- and post-weathered samples.

Research Experience

Edge Computing and UAV Research | Supervisor: Prof. Bo Xiao | 2024.10 — Present

[Submitted] Yihui Yan, Pengfei Ren, et al., *MAO-LCR: A Multi-Agent Learning Approach for Joint Task Offloading and Cache Replacement in Edge Computing*, *IEEE WCNC 2026*

Proposed a regret-aware multi-agent reinforcement learning framework to jointly optimize task offloading and caching strategies, effectively reducing latency and energy consumption while improving cache hit rate.

[Under Review] Yihui Yan, Huimin Cao, et al., *Multi-Agent Proximal Policy Optimization for Joint Task Offloading and Trajectory Control in NOMA-based Multi-UAV MEC Networks*, *IEEE Internet of Things Journal (IoTJ)*

Developed a multi-agent reinforcement learning approach for joint task offloading and trajectory optimization in dynamic NOMA-MEC networks to minimize system delay and energy consumption.

Computer Vision Research | Supervisor: Prof. Liang Zhang | 2022.02 — 2023.06

[Accepted] Yihui Yan: *AlexViT: Novel Diabetic Retinopathy Image Classification*, *IEEE ICETCI 2023*

Proposed a lightweight hybrid model combining the global attention of **Vision Transformer (ViT)** with the local feature extraction of **AlexNet**. The model introduces a convolutional embedding and attention fusion mechanism, achieving a **45% parameter reduction** and a **3.7% improvement in Kappa score**, attaining state-of-the-art performance on EyePACS and Messidor datasets.

Languages & Skills

- **Languages:** CET-4 (587), CET-6 (530)
- **Programming:** Java, Python, C++
- **Technical Tools:** Git workflow, L^AT_EX, Markdown, Simulink, Arduino, Matlab

Student Activities

Communist Party member; Core member of the Cultural Department, Youth League Committee of Xidian University; Director of Human Resources Department, Graduate Student Union, ECNU; Class Development Committee member.

Personal Summary

Outstanding academic performance with a strong foundation in computer science and artificial intelligence. Highly self-motivated, able to independently learn and apply new technologies. Skilled in bridging research and engineering—capable of deploying cutting-edge algorithms into real-world systems. Experienced in project implementation, emphasizing teamwork, reliability, and practical impact.